

The minimum order of a counterexample to a conjecture on integral generalized sun graphs

Horace Dong

Independent researcher

maxwelldhx@gmail.com

ORCID: [0009-0002-7554-7548](https://orcid.org/0009-0002-7554-7548)

September 2026

Abstract

A graph is integral if its adjacency eigenvalues are integers, and a generalized sun graph is a cycle with pendant paths attached to its vertices. Braga, Del-Vecchio and Rodrigues conjectured that the cycle of an integral generalized sun graph that is not a cycle has length divisible by 4. Braga, Moraes and Santos disproved this with the graph $C_{6,1}(0, 6, 6, 12, 6, 6)$ on 42 vertices, and remarked that their computer search does not show that no smaller counterexample exists. We show that none exists: up to isomorphism, $C_{6,1}(0, 6, 6, 12, 6, 6)$ is the only counterexample with at most 42 vertices, also when pendant paths of both admissible lengths share a vertex. Every integral generalized sun graph with at most 41 vertices that is not a cycle has a cycle of length 4 and is one of the three graphs found by Braga, Rodrigues and Trevisan in 2017. Up to 42 vertices, spectral lemmas exclude every cycle length $b \not\equiv 0 \pmod{4}$ except 3, 6 and 10, which are settled by exhaustive searches with separately written programs, two of them by independent referee agents. We also note that if an integral unicyclic graph has a cycle of odd length b , then every prime q with $(q-1) \mid (b-1)$ divides $2b$; hence $3 \mid b$, and $b = 3$ or $b \geq 15$. Nothing in this paper is formally verified. The results were found by AI coding agents working under the author's direction; their role is described at the end of the paper.

1 Introduction

A graph is *integral* if all eigenvalues of its adjacency matrix are integers, a notion that goes back to Harary and Schwenk [6]. A search of Braga, Rodrigues and Trevisan [3] showed that, besides the cycles C_3 , C_4 and C_6 , exactly three unicyclic graphs with at most 21 vertices are integral. Braga, Del-Vecchio and Rodrigues [1] constructed infinite families of integral unicyclic graphs by attaching pendant paths to the vertices of a cycle; such graphs are called *generalized sun graphs*. Following [2], $C_b(n_1P_{t_1}, \dots, n_bP_{t_b})$ is obtained from the cycle $v_1 \cdots v_b v_1$ by attaching n_k copies of the path P_{t_k} on t_k vertices to v_k (joining v_k to an end of each copy), and $C_{b,t}(n_1, \dots, n_b)$ is the case $t_1 = \cdots = t_b = t$. We state the second conjecture of [1] in our words.

Conjecture 2 ([1]; see [2, Conjecture 2]). *Let G be an integral generalized sun graph that is not a cycle. Then the cycle of G has length divisible by 4.*

Braga, Moraes and Santos [2] proved the first conjecture of [1] in a strong form: a graph with a pendant path of at least three edges has an eigenvalue in $(1, 2\cos(\pi/9)]$, so it is not integral [2, Theorem 2.1]. They disproved Conjecture 2: $C_{6,1}(0, 6, 6, 12, 6, 6)$ has 42 vertices and spectrum $\{0^{[32]}, \pm 2^{[2]}, \pm 3^{[2]}, \pm 4\}$ [2, Proposition 3.2], and it is the smallest member of an infinite family [2, Corollary 3.6 and the remark after it]. Their search over boxes of parameter vectors, for graphs with pendant vertices only, found no other integral graph whose cycle length

is not divisible by 4. As they point out [2, Section 4], it “does not prove that the graph on 42 vertices has minimum order” among all counterexamples, since vectors outside the boxes may give smaller graphs, and graphs with copies of P_2 were not searched. This note answers that question.

A *counterexample* is a generalized sun graph that is integral, is not a cycle, and has cycle length $b \not\equiv 0 \pmod{4}$. We allow any finite number of pendant paths at each cycle vertex, possibly of different lengths at the same vertex. By [2, Theorem 2.1], an integral generalized sun graph then carries only copies of P_1 and P_2 . The graphs $C_b(n_1P_{t_1}, \dots, n_bP_{t_b})$ form a subclass containing every graph named below, so our results hold in either reading.

Theorem 1. *Up to isomorphism, $C_{6,1}(0, 6, 6, 12, 6, 6)$ is the only counterexample to Conjecture 2 with at most 42 vertices. In particular, the minimum order of a counterexample is 42.*

Theorem 2. *Every integral generalized sun graph with at most 41 vertices that is not a cycle has a cycle of length 4, and is isomorphic to $C_{4,1}(6, 0, 3, 0)$, with 13 vertices, or to $C_4(5P_2, P_1, 2P_2, P_1)$ or $C_{4,2}(4, 4, 0, 0)$, with 20 vertices each.*

These three graphs are the three integral unicyclic graphs found by Braga, Rodrigues and Trevisan [3, Section 5, p. 296, and Table 1, p. 297], with spectra $\{0^{[9]}, \pm 2, \pm 3\}$ and, for the two cospectral graphs, $\{0^{[2]}, \pm 1^{[7]}, \pm 2, \pm 3\}$. Theorem 2 adds that there is no other integral generalized sun graph up to 41 vertices. The one-page abstract of Veras [13] reports a computer search for integral unicyclic graphs. It names $C_{4,1}(0, 16, 32, 16)$, on 68 vertices, and the family $C_{4,1}(0, a, 2a, a)$, whose integral members have at least 68 vertices [2]. It does not state the range of the search and names no graph with at most 41 vertices, so nothing it reports overlaps Theorem 2, although the search may have covered part of that range. Write $\text{mult}(\lambda)$ for the multiplicity of an eigenvalue λ . The proofs use the following observation on integral graphs.

Proposition 3. *Let G be an integral graph with e edges that contains an odd cycle, let g be the length of its shortest odd cycles and c_g their number. Then every prime q with $(q-1) \mid (g-1)$ divides $2gc_g$; in particular $3 \mid gc_g$. Moreover, at least $(g-1)/2$ integers $v \geq 2$ satisfy $\text{mult}(v) \neq \text{mult}(-v)$, so $2e \geq \sum_{v=2}^{(g+1)/2} v^2$.*

For an integral unicyclic graph with a cycle of odd length b this gives $3 \mid b$, and $b = 3$ or $b \geq 15$, with at least 102 vertices in the latter case (Corollary 3.1).

Method. Up to 42 vertices, hand proofs exclude odd $b \geq 5$ (Proposition 3), $b \in \{14, 18\}$ (Corollary 4.3) and even $b \geq 22$ (Lemma 4.1), and a bound due to the first referee agent (see the disclosure; Lemma 4.4) excludes every $b \geq 14$ again; exhaustive searches settle $b = 3, 6, 10$ (Section 5). Every case is confirmed at least twice, by hand proofs or by separately written programs (Table 3); two of the programs, C and D, were written from scratch by referee agents that had not produced the results. Nothing here is formally verified, in Lean or otherwise; the results rest on the written proofs and the computations.

Prior work. The question was raised in version 1 of [2], submitted to the arXiv on 23 September 2026. We found no earlier answer to it and no discussion of it. Our searches on 26 September 2026, 18:58–19:43 UTC, covered the arXiv listing of [2] (version 1 only); Crossref; MathDB (<https://mathdb.com>); GitHub issues, pull requests, commits, code and repositories, including the `trureturing` repository [11] of AI-assisted work on open problems (as a positive control, the same search for the versioned identifier of another recent arXiv paper found the expected issue); MathOverflow and Mathematics Stack Exchange; and web search. The OpenAlex search and citation queries failed with HTTP error 429 (rate limit), and several GitHub searches were rate-limited and repeated later. From 19:59 to 20:32 UTC the first referee agent repeated part of these searches and added arXiv API queries and full-text

searches (Remark 3.2). We did not search Google Scholar, zbMATH or MathSciNet. The earlier computer searches for integral unicyclic graphs of [3] and [13] are discussed after Theorem 2; we read the abstract [13], which [2] cite, on 26 September 2026 at 22:48 UTC, and at 22:56 UTC the arXiv still listed only version 1 of [2]. A final check on 26 September 2026 between 23:15 and 23:17 UTC (the arXiv, GitHub, MathDB and MathOverflow) found nothing new. This reports the coverage of our searches, not a guarantee of priority.

2 Notation and spectral lemmas

Throughout, G is a generalized sun graph with cycle $v_0v_1 \cdots v_{b-1}v_0$ ($b \geq 3$, indices modulo b) carrying only copies of P_1 and P_2 : at v_k hang p_k pendant vertices and q_k pendant paths v_kww' . Put $P = \sum p_k$, $Q = \sum q_k$, $K_p = \{k : p_k > 0\}$, $K_q = \{k : q_k > 0\}$, $Z = \mathbb{Z}_b \setminus K_q$ and $A = A(G)$. Then G has $n = b + P + 2Q$ vertices and n edges. Let $\Theta(G) = \sum_v d_v(d_v - 1)$ over all vertices, with degrees d_v ; then $\Theta(G) = \sum_k ((2 + p_k + q_k)(1 + p_k + q_k) + 2q_k)$. An isomorphism maps the unique cycle onto the unique cycle, and (p_k, q_k) is determined by the tree attached at v_k , so isomorphism classes correspond to dihedral orbits of the sequences $((p_k, q_k))_{k \in \mathbb{Z}_b}$.

For a real symmetric matrix M , $n_+(M)$, $n_-(M)$ and $n_0(M)$ count its positive, negative and zero eigenvalues. For a graph H , $N_{>t}(H)$, $N_{\geq t}(H)$, $N_{<t}(H)$ and $N_{\leq t}(H)$ count the eigenvalues of $A(H)$ in the indicated ranges, with multiplicity. If G is integral, B^+ is the multiset of its eigenvalues $\lambda \geq 2$ and $S_2 = \sum_{\lambda \in B^+} \lambda^2$. For nonempty $K = \{k_1 < \cdots < k_t\} \subseteq \{0, \dots, b-1\}$ put $k_{t+1} = k_1 + b$; the *gaps* of K are the pairs (k_i, k_{i+1}) , of *length* $k_{i+1} - k_i$. If $|K| = 1$, the only gap has length b .

Lemma 2.1. *For $\lambda \in \mathbb{R} \setminus \{-1, 0, 1\}$ let $f_k(\lambda) = p_k/\lambda + q_k\lambda/(\lambda^2 - 1)$ and $S(\lambda) = A(C_b) - \lambda I + \text{diag}(f_k(\lambda))_k$.*

- (a) *If $\lambda > 1$, then $N_{>\lambda}(G) = n_+(S(\lambda))$ and $N_{\geq \lambda}(G) = n_+(S(\lambda)) + n_0(S(\lambda))$.*
- (b) *If $\lambda < -1$, then $N_{<\lambda}(G) = n_-(S(\lambda))$ and $N_{\leq \lambda}(G) = n_-(S(\lambda)) + n_0(S(\lambda))$.*
- (c) *$N_{>1}(G) \geq |K_q| + n_+(R)$ and $N_{<-1}(G) \geq |K_q| + n_-(R')$, where R and R' are the principal submatrices of $A(C_b) - I + \text{diag}(p_k)$ and $A(C_b) + I - \text{diag}(p_k)$ on Z .*

Proof. With the cycle first, $A - \lambda I = \begin{pmatrix} A(C_b) - \lambda I & W \\ W^\top & H \end{pmatrix}$, where $H = H(\lambda)$ is the direct sum of a block $(-\lambda)$ for each pendant vertex and a block $\begin{pmatrix} -\lambda & 1 \\ 1 & -\lambda \end{pmatrix}$ for each pendant path (rows w, w'), and W has one entry 1 in the column of each pendant vertex and each w , in the row of its neighbour on the cycle. H has eigenvalues $-\lambda$, $-\lambda \pm 1$, so it is negative (positive) definite for $\lambda > 1$ ($\lambda < -1$). The 2×2 block has inverse with $(1, 1)$ entry $-\lambda/(\lambda^2 - 1)$, so $WH^{-1}W^\top = -\text{diag}(f_k(\lambda))$ and $A - \lambda I$ is congruent to $S(\lambda) \oplus H$; now use Sylvester's law of inertia.

(c) As $\lambda \downarrow 1$, the entries $f_k(\lambda)$, $k \in K_q$, tend to $+\infty$ while the other entries of $S(\lambda)$ stay bounded; so the block S_{KK} , $K = K_q$, is positive definite for λ near 1 and its inverse tends to 0. By Haynsworth's inertia additivity [7], $n_+(S(\lambda)) = |K| + n_+(S_{ZZ} - S_{ZK}S_{KK}^{-1}S_{KZ})$, and the Schur complement tends to R , so its positive inertia is eventually at least $n_+(R)$. Finally $N_{>1}(G) = N_{>\lambda}(G)$ for $\lambda > 1$ close to 1. The case $\lambda \uparrow -1$ is the same, with $f_k(\lambda) \rightarrow -\infty$ on K_q . $\square$

Lemma 2.2. (a) *Let $m \in \mathbb{R} \setminus \{-1, 0, 1\}$, $c_k = m - f_k(m)$, $T_k = \begin{pmatrix} c_k & -1 \\ 1 & 0 \end{pmatrix}$ and $M = T_{b-1} \cdots T_0$. Then $\text{mult}(m) = \dim \ker(M - I) \leq 2$; it equals 2 if and only if $M = I$, and 1 if and only if $M \neq I$ and $\text{tr } M = 2$.*

- (b) *$\text{mult}(0) = P - |K_p| + \delta_0$, where, if $K_p = \emptyset$, $\delta_0 = 2$ when $4 \mid b$ and $\delta_0 = 0$ otherwise, and, if $K_p \neq \emptyset$, δ_0 is the number of gaps of K_p of even length.*

- (c) For $\varepsilon = \pm 1$, $\text{mult}(\varepsilon) = Q - |K_q| + \delta_\varepsilon$, where δ_ε is the dimension of the space of $y \in \mathbb{R}^{\mathbb{Z}_b}$ with $y_k = 0$ for $k \in K_q$ and $y_{k-1} + y_{k+1} = \varepsilon(1 - p_k)y_k$ for $k \notin K_q$. Each gap of K_q contributes 0 or 1 to δ_ε , and $\delta_\varepsilon \leq 2$ if $K_q = \emptyset$.
- (d) $\text{mult}(0) \leq \max(P, 2)$ and $\text{mult}(\pm 1) \leq \max(Q, 2)$.

Proof. (a) If $Ax = mx$, then $x_a = x_{v_k}/m$ at a pendant vertex a of v_k , and $x_{w'} = x_w/m$, $x_w = mx_{v_k}/(m^2 - 1)$ on a pendant path $v_kw w'$. So x is determined by its restriction y to the cycle, which satisfies $y_{k-1} + y_{k+1} = c_k y_k$, and every solution y extends to an eigenvector. Writing $(y_{k+1}, y_k)^\top = T_k(y_k, y_{k-1})^\top$, the solutions correspond to the vectors fixed by M ; as $\det M = 1$, $M - I$ is singular if and only if $\text{tr } M = 2$.

(b) If $Ax = 0$, a pendant vertex at v_k forces $x_{v_k} = 0$, and a pendant path gives $x_w = 0$, $x_{w'} = -x_{v_k}$. For $k \in K_p$ the equation at v_k only fixes the sum of the p_k pendant values ($p_k - 1$ free values), and for $k \notin K_p$ it reads $y_{k-1} + y_{k+1} = 0$. So δ_0 is the dimension of the space of y with $y_k = 0$ on K_p and $y_{k-1} + y_{k+1} = 0$ off K_p . If $K_p = \emptyset$, then $y_{k+2} = -y_k$ for all k , with a 2-dimensional space of b -periodic solutions if $4 \mid b$ and none otherwise. If $K_p \neq \emptyset$, the equations split along the gaps. On a gap of length d put $z_j = y_{k_i+j}$; then $z_0 = z_d = 0$ and $z_{j-1} + z_{j+1} = 0$ for $0 < j < d$, which forces $z_j = 0$ for even j and $z_{j+2} = -z_j$ for odd j . For even d this leaves z_1 free; for odd d it gives $z_1 = \pm z_d = 0$.

(c) If $Ax = \varepsilon x$, a pendant path gives $x_{w'} = \varepsilon x_w$ and $\varepsilon x_w = x_{v_k} + x_{w'}$, so $x_{v_k} = 0$ for $k \in K_q$, and the equation at v_k only fixes the sum of the q_k values x_w ($q_k - 1$ free values). A pendant vertex gives $x_a = \varepsilon x_{v_k}$, so for $k \notin K_q$ the equation at v_k reads $y_{k-1} + y_{k+1} + \varepsilon p_k y_k = \varepsilon y_k$. On a gap of K_q the recurrence determines all values from the first one after the left end, so the gap contributes at most 1; if $K_q = \emptyset$ the periodic solutions of a recurrence of order two form a space of dimension at most 2.

(d) If $K_p \neq \emptyset$, then $\delta_0 \leq |K_p|$, the number of gaps, so $\text{mult}(0) \leq P$; otherwise $\text{mult}(0) = \delta_0 \leq 2$. Likewise for ± 1 . $\square$

Lemma 2.3. *Let H be a graph with e edges, degrees d_v , c_4 cycles of length 4, and adjacency matrix A .*

$$(a) \text{tr } A^2 = 2e \text{ and } \text{tr } A^4 = 2e + 2 \sum_v d_v(d_v - 1) + 8c_4.$$

(b) *If the shortest odd cycles of H have length g and there are c_g of them, then $\text{tr } A^L = 0$ for odd $L < g$ and $\text{tr } A^g = 2g c_g$.*

Proof. We count closed walks. (a) A closed walk of length 4 traverses one edge four times (2 walks per edge), or two edges with a common vertex twice each (4 walks for each of the $\sum_v \binom{d_v}{2}$ such pairs), or a 4-cycle (8 walks per cycle). (b) A closed walk of odd length L that repeats a vertex splits there into two shorter closed walks, one of odd length. By induction it contains an odd cycle of length at most L , with equality only if it traverses a cycle once. So no closed walk of odd length $L < g$ exists, and each g -cycle gives $2g$ closed walks of length g . $\square$

For our graphs $e = n$, $c_4 = [b = 4]$, $\sum_v d_v(d_v - 1) = \Theta(G)$, and $g = b$, $c_g = 1$ if b is odd. If G is integral, the eigenvalues in $\{0, \pm 1\}$ satisfy $\lambda^4 = \lambda^2$ and $\lambda^L = \lambda$ for odd L , and $\text{tr } A = 0$, so

$$\sum_\lambda \lambda^2 = 2n, \quad \sum_{|\lambda| \geq 2} \lambda^2(\lambda^2 - 1) = 2\Theta(G) + 8[b = 4], \quad (1)$$

$$\sum_{|\lambda| \geq 2} (\lambda^L - \lambda) = 0 \quad (L \text{ odd}, 3 \leq L < b), \quad \sum_{|\lambda| \geq 2} (\lambda^b - \lambda) = 2b \text{ if } b \text{ is odd.} \quad (2)$$

Lemma 2.4. *Let G be integral and not a cycle.*

(a) *The largest eigenvalue ρ is simple and $\rho \geq 3$; every eigenvalue satisfies $|\lambda| \leq \rho$ and $|\lambda| \leq \sqrt{2n}$, and $\lambda > -\rho$ if b is odd.*

(b) *$\text{mult}(\lambda) \leq 2$ whenever $|\lambda| \geq 2$.*

(c) If b is even, the spectrum is symmetric, $S_2 = n - \text{mult}(1)$ and $P + Q \leq S_2 - b + 2$. If b is odd, $P + Q \leq \frac{1}{2} \sum_{|\lambda| \geq 2} \lambda^2 - b + 2$.

(d) If b is even, then $|B^+| \leq 4$ if $n \leq 41$, and $|B^+| \leq 5$ if $n \leq 42$, with equality only if $B^+ = \{4, 3, 3, 2, 2\}$.

Proof. (a) By Perron–Frobenius theory [4], ρ is simple, $|\lambda| \leq \rho$, and $-\rho$ is an eigenvalue only if G is bipartite. As G properly contains C_b , $\rho > 2$. Also $\lambda^2 \leq \text{tr } A^2 = 2n$. (b) is Lemma 2.2(a). (c) For even b , G is bipartite and (1) gives $2n = 2S_2 + 2 \text{mult}(1)$; by Lemma 2.2(d), $S_2 \geq b + P + 2Q - \max(Q, 2) \geq b + P + Q - 2$. For odd b , $\sum_{|\lambda| \geq 2} \lambda^2 = 2n - \text{mult}(1) - \text{mult}(-1)$, and the same estimate applies. (d) By (a) and (b), B^+ has a simple largest element and contains every other value at most twice. Five elements need $\rho \geq 4$ and $S_2 \geq 4^2 + 2 \cdot 3^2 + 2 \cdot 2^2 = 42$, with equality only for $\{4, 3, 3, 2, 2\}$, and six need $S_2 \geq 5^2 + 4^2 + 2 \cdot 3^2 + 2 \cdot 2^2 = 67$; and $S_2 \leq n$ by (c). $\square$

3 Odd cycles

Proof of Proposition 3. For $v \geq 2$ let $\eta_v = \text{mult}(v) - \text{mult}(-v)$. For odd L the eigenvalues in $\{0, \pm 1\}$ satisfy $\lambda^L = \lambda$, and $(-v)^L - (-v) = -(v^L - v)$, so by Lemma 2.3(b) and $\text{tr } A = 0$,

$$\sum_{v \geq 2} \eta_v (v^L - v) = \text{tr } A^L = 0 \quad (L \text{ odd}, L < g), \quad \sum_{v \geq 2} \eta_v (v^g - v) = 2g c_g. \quad (3)$$

Let q be prime with $(q-1) \mid (g-1)$. For every integer λ , $\lambda^g \equiv \lambda \pmod{q}$: this is clear if $q \mid \lambda$, and otherwise $\lambda^{q-1} \equiv 1$ by Fermat's little theorem, so $\lambda^g = \lambda(\lambda^{q-1})^{(g-1)/(q-1)} \equiv \lambda$. Hence q divides $\sum_{\lambda} (\lambda^g - \lambda) = 2g c_g$. The prime $q = 3$ qualifies as g is odd.

Let $V = \{v \geq 2 : \eta_v \neq 0\}$, $s = |V|$, and suppose $s \leq (g-3)/2$. The exponents $L = 2j+1$, $1 \leq j \leq s$, are odd and less than g , so $\sum_{v \in V} \eta_v v(w_v^j - 1) = 0$ for $1 \leq j \leq s$, with $w_v = v^2$. Subtracting the row of the value 1 from the other rows of the Vandermonde matrix of the distinct numbers 1 and w_v ($v \in V$) shows that $\det(w_v^j - 1)$ equals that Vandermonde determinant, which is nonzero. Hence all $\eta_v = 0$, which contradicts the definition of V if $s > 0$ and (3) if $s = 0$. So $s \geq (g-1)/2$, and $2e = \sum_{\lambda} \lambda^2 \geq \sum_{v \in V} v^2 \geq \sum_{v=2}^{(g+1)/2} v^2$. $\square$

Corollary 3.1. *Let G be an integral unicyclic graph with n vertices whose cycle has odd length b , and let κ_b be the product of the primes q with $(q-1) \mid (b-1)$. Then $\kappa_b \mid 2b$; in particular $3 \mid b$. The odd $b < 200$ with $\kappa_b \mid 2b$ are 3, 15, 27, 39, 63, 75, 87, 99, 123, 135, 147, 159, 183 and 195. If $b \neq 3$, then $b \geq 15$ and $n \geq 102$.*

Proof. Here $g = b$, $c_g = 1$ and $e = n$, so $\kappa_b \mid 2b$ by Proposition 3. For $b = 5, 7, 9, 11, 13$, $\kappa_b = 30, 42, 30, 66, 2730$ does not divide $2b$; the list is a direct computation. If $b \geq 15$, then $2n \geq \sum_{v=2}^8 v^2 = 203$. $\square$

Remark 3.2. By the theorem of von Staudt and Clausen, κ_b is the denominator of the Bernoulli number B_{b-1} . The first part of Proposition 3 uses integrality; for an arbitrary integer matrix one only has the classical congruence $\text{tr } A^q \equiv \text{tr } A \pmod{q}$ for primes q . The second part is a variant of the odd-polynomial argument of van Dam and Haemers [12, Section 2]. We found no earlier statement of Proposition 3, even for unicyclic graphs (20 web searches, 9 arXiv API queries and full-text searches of related papers), but the paper of Omid $\text{\textcircled{O}}$ [9] on necessary conditions for integral unicyclic graphs and [1] could not be read in full (paywalled). As the argument is elementary, its novelty is uncertain; we present it as an observation.

4 Restricting the cycle length

Lemma 4.1. $N_{>1}(G) \geq N_{>1}(C_b) = 2\lceil b/6 \rceil - 1$ and $N_{<-1}(G) \geq N_{<-1}(C_b)$. So if G is integral and not a cycle, then $|B^+| \geq 2\lceil b/6 \rceil - 1$; if moreover b is even, then $b \leq 12$ when $n \leq 41$, and $b \leq 18$ when $n \leq 42$, with $B^+ = \{4, 3, 3, 2, 2\}$ for $b \in \{14, 16, 18\}$. In particular, if $n \leq 42$ and $b \equiv 2 \pmod{4}$, then $b \in \{6, 10, 14, 18\}$.

Proof. C_b is an induced subgraph of G , so by Cauchy interlacing [4] the i -th largest (smallest) eigenvalue of G is at least (at most) that of C_b for $i \leq b$. The eigenvalues of C_b are $2\cos(2\pi j/b)$, $j \in \mathbb{Z}_b$, and $2\cos(2\pi j/b) > 1$ if and only if $6|j| < b$ for $j \in (-b/2, b/2]$, which holds for $2\lceil b/6 \rceil - 1$ values of j . If G is integral, its eigenvalues greater than 1 form B^+ . Now apply Lemma 2.4(d); $2\lceil b/6 \rceil - 1 = 5$ for $13 \leq b \leq 18$. $\square$

Lemma 4.2. Let G be integral with b even, and $s = |B^+|$. Then $\text{mult}(\pm 1) = n - S_2$ and $\text{mult}(0) = 2S_2 - n - 2s$. Hence $S_2 \leq n \leq 2S_2 - 2s$ and $2S_2 - n - 2s \leq \max(P, 2)$.

Proof. By symmetry $n = \text{mult}(0) + 2\text{mult}(1) + 2s$, and (1) gives $2n = 2S_2 + 2\text{mult}(1)$. The last inequality is Lemma 2.2(d). $\square$

Corollary 4.3. No integral generalized sun graph with cycle length 14 or 18 has at most 42 vertices.

Proof. By Lemma 4.1, $s = 5$ and $S_2 = 42$. By Lemma 4.2, $42 \leq n \leq 42$, so $\text{mult}(1) = 0$ and $\text{mult}(0) = 84 - 42 - 10 = 32 \leq \max(P, 2)$. Then $n \geq b + P \geq 46$, a contradiction. $\square$

The next bound is due to the first referee agent (see the disclosure) and is the pruning rule of implementation C in Section 5. For $t \geq 1$ let $\sigma(t)$ be the sum of the first t terms of $4, 4, 4, 4, 9, 9, 9, 9, 16, \dots$, in which each v^2 , $v \geq 2$, appears four times, and let $\sigma(t) = 0$ for $t \leq 0$. Thus $\sigma(9) = 68$, $\sigma(10) = 84$, $\sigma(11) = 100$, $\sigma(14) = 166$.

Lemma 4.4. Let G be integral and not a cycle, r the number of eigenvalues with $|\lambda| \geq 2$, and $r_0 = b + |K_p| - \delta_0$, with $\delta_0, \delta_{\pm 1}$ as in Lemma 2.2. Then

- (a) $r = r_0 + 2|K_q| - \delta_1 - \delta_{-1} \geq b - 4$;
- (b) $\sigma(r) \leq 2n$;
- (c) $\sigma(r_0) \leq 2n - 2|K_q|$ if $K_q \neq \emptyset$, and $\sigma(r_0 - 4) \leq 2n - 4$ if $K_q = \emptyset$.

Consequently, $b \leq 13$ if $n \leq 42$.

Proof. (a) Since $r = n - \text{mult}(0) - \text{mult}(1) - \text{mult}(-1)$, the formula follows from Lemma 2.2(b),(c). If $K_p \neq \emptyset$, then $\delta_0 \leq |K_p|$ and $r_0 \geq b$; otherwise $K_q \neq \emptyset$ and $r_0 \geq b - 2$. With $\delta_{\pm 1} \leq |K_q|$ if $K_q \neq \emptyset$ and $\delta_{\pm 1} \leq 2$ otherwise, $r \geq b - 4$. (b) Each value $\pm v$, $v \geq 2$, occurs at most twice (Lemma 2.4(b)), so the squares of these r eigenvalues sum to at least $\sigma(r)$ and at most $2n$. (c) If $K_q \neq \emptyset$, put $x = 2|K_q| - \delta_1 - \delta_{-1} \geq 0$; then $r = r_0 + x$ and $\text{mult}(1) + \text{mult}(-1) = 2Q - x$, so $\sigma(r_0) + 4x \leq \sigma(r) \leq 2n - 2Q + x$, whence $\sigma(r_0) \leq 2n - 2|K_q|$. If $K_q = \emptyset$, put $y = \delta_1 + \delta_{-1} \leq 4$; then $r = r_0 - y$ and $\sigma(r_0 - 4) + 4(4 - y) \leq \sigma(r_0 - y) \leq 2n - y$ (trivially if $r_0 < 4$), whence $\sigma(r_0 - 4) \leq 2n - 4$.

Let $n \leq 42$. If $b \geq 15$, (a) and (b) give $100 = \sigma(11) \leq 84$. If $b = 14$, then $r_0 \geq 14$ (as $4 \nmid 14$), and (c) gives $166 \leq 82$ or $84 \leq 80$. $\square$

5 Computations

5.1 The programs

Each program takes b and N , runs through the sequences $((p_k, q_k))_k$, not all zero, with $n \leq N$, allowing $p_k, q_k > 0$ at the same vertex, and decides exactly which graphs are integral. Graphs reported as integral were confirmed by factoring the characteristic polynomial with SymPy [8].

Implementation A (**sunsearch.c**) lists the admissible spectra (Lemma 2.4 and, for odd b , (2); version 5 adds Lemma 4.1, version 6 optionally Lemma 4.2) and assigns (p_k, q_k) to $v_0, v_1, \dots$ by depth-first search, with a lexicographically largest pair at v_0 . A node is discarded if no admissible spectrum is compatible with the bounds of Lemma 5.1, (1) and Lemma 2.4(c); as these bounds are monotone, the loops over p_k and q_k stop at the first discarded value. With Lemma 4.2 switched on, version 6 also uses its inequalities in this test, with $n \geq b + P + 2Q$ for the partial sums P and Q ; they are monotone as well. The window of Lemma 5.3 only skips a node, and the last pair is solved from (1). At a leaf, A computes $\text{mult}(m)$ for all $|m| \leq 9$ by Lemma 2.2, for $|m| \geq 2$ with the integer matrices hT_k , $h = m(m^2 - 1)$, modulo four primes near 2^{62} and with an a priori bound below 2^{240} that makes the tests exact; G is integral if and only if these multiplicities add up to n , as $|\lambda| \leq \sqrt{2n} < 10$.

Implementation B (**sunB.c**) was written separately, by the same agent. Version 1 prunes only by the largest admissible value of $\Theta(G)$, tests (1) at the leaves (using $\text{mult}(\pm 1) \leq \max(Q, 2)$), and then tests the dihedrally canonical sequences (least image) exactly, via

$$F(x) = \det(x(x^2 - 1)(xI - A(C_b)) - \text{diag}(p_k(x^2 - 1) + q_k x^2)) = x^{b-P}(x^2 - 1)^{b-Q} \det(xI - A),$$

the equality following from the proof of Lemma 2.1. F is monic of degree $4b$, and if G is integral its roots are integers in $[-9, 9]$. B computes F modulo three primes by interpolation; if the multiplicities of the roots $m \in [-9, 9]$ modulo some prime add up to less than $4b$, then G is not integral. Version 2 adds Lemma 4.1 and the window of Lemma 5.3; version 3 adds Lemma 4.2, the rotation rule of A and the greatest image as canonical form. B2 (**sunB2.c**) is B (version 1) with the rotation rule of A, the greatest image as canonical form and a floating-point interlacing test (LAPACK's **dsyev**, margin 10^{-6}) on an induced subgraph of every completion; version 2 adds Lemma 4.1 and takes a larger induced subgraph. B2 serves only as a cross-check. (The header comments of **sunB.c**, version 3, and of both versions of B2 still describe the least image.)

Implementation C (**sunC.c**, **sunC2.c**) was written from scratch by the first referee agent. It enumerates support patterns (whether $p_k > 0$, whether $q_k > 0$) up to dihedral symmetry and then all values on each pattern. It discards patterns by Lemma 4.4(a),(b) (C), also by Lemma 4.4(c) (C2), or not at all. Its exact test computes $\phi(G) = \det(xI - A)$ modulo two primes by Schwenk's formula [10] for the edge $e = v_{b-1}v_0$, $\phi(G) = \phi(G - e) - \phi(G - v_{b-1} - v_0) - 2\phi(G - V(C_b))$, and, as B does, adds up the root multiplicities of the integers in $[-10, 10]$ modulo each prime, which must give n if G is integral.

Implementation D (**sunD_v1.c**, **sunD2.c**, **sunD3.c**) was written from scratch by a second referee agent (see the disclosure). It runs through all sequences, with no rotation rule, or only through those that are canonical for the greatest image under the key $64(p_k + 2q_k) + p_k$. Its only pruning, for even b , is exact reachability, computed by dynamic programming: some completion and some multiset B^+ with a simple largest element $\rho \geq 3$ and other elements in $[2, \rho - 1]$, each at most twice, must satisfy the second identity of (1) and $S_2 \leq n \leq \min(N, S_2 + \max(Q, 2))$ (version 1 omits the bound $S_2 + \max(Q, 2)$). Version 3 can switch the pruning off, and then also accepts odd b . For each integer $k \in [-9, 9]$, D bounds $\text{mult}(k)$ from above by the nullity modulo $2^{31} - 1$ of a $b \times b$ integer matrix: $k(k^2 - 1)S(k)$ for $|k| \geq 2$ (Lemma 2.1), and the systems of Lemma 2.2(b),(c) for $k \in \{0, \pm 1\}$. If these bounds add up to less than n , then G is not integral. The runs of D used no inertia bound and none of Lemmas 4.1–4.4 or 5.3.

5.2 Soundness of the pruning

Let $X = \{0, \dots, b-2\}$. Below, a tridiagonal matrix indexed by $Y \subseteq X$ has off-diagonal entry 1 in the positions $(k, k+1)$ and $(k+1, k)$ for which both k and $k+1$ lie in Y , and 0 elsewhere.

Lemma 5.1. *Let $m \geq 2$ and let T be tridiagonal on X with diagonal entries $t_k \leq f_k(m) - m$. Then $N_{>m}(G), N_{<-m}(G) \geq n_+(T)$ and $N_{\geq m}(G), N_{\leq -m}(G) \geq n_+(T) + n_0(T)$. If $Y \subseteq K_q \cap X$ and T' is tridiagonal on $X \setminus Y$ with diagonal entries $t'_k \leq p_k - 1$, then $N_{>1}(G), N_{<-1}(G) \geq |Y| + n_+(T')$.*

Proof. The principal submatrix $S(m)_X$ is tridiagonal, as $b-1 \notin X$, and $S(m)_X - T$ is diagonal and nonnegative. By Weyl's inequalities and Cauchy interlacing [4], $n_+(S(m)) \geq n_+(S(m)_X) \geq n_+(T)$, and likewise for $n_+ + n_0$; apply Lemma 2.1(a). As f_k is odd, $\Delta S(-m)_X \Delta = -S(m)_X$ for $\Delta = \text{diag}((-1)^k)$, and Lemma 2.1(b) gives the negative side. For T' , let $U = (K_q \cap X) \setminus Y$. By Lemma 2.1(c) and interlacing, $N_{>1}(G) \geq |K_q| + n_+(R_{Z \cap X})$; and $R_{Z \cap X}$ arises from the tridiagonal matrix on $X \setminus Y$ with diagonal $p_k - 1$ by deleting the indices in U , so $n_+(R_{Z \cap X}) \geq n_+(T') - |U|$, while $|K_q| \geq |Y| + |U|$. The negative side is the same with R' and Δ . $\square$

At a node of A with $v_0, \dots, v_{d-1}$ assigned ($d \leq b-1$), T and T' use the assigned values and, elsewhere, those of a bare vertex ($t_k = -m, t'_k = -1$), with $Y = K_q \cap \{0, \dots, d-1\}$. Raising p_k or q_k raises t_k, t'_k or moves k into Y , which does not decrease $|Y| + n_+(T')$; so the bounds are monotone. Inertias are computed by the pivot recurrence $d_1 = t_1, d_k = t_k - 1/d_{k-1}$, restarted after each index missing from the index set, in exact rational arithmetic (no overflow occurred).

Lemma 5.2. *Let T be symmetric tridiagonal with nonzero off-diagonal entries. If an exact zero pivot $d_k = 0$ is counted as negative, d_{k+1} as positive, and the recurrence is restarted with $d_{k+2} = t_{k+2}$, then the number of positive pivots is $n_+(T)$; with the signs exchanged one obtains $n_-(T)$.*

Proof. For small $\varepsilon > 0$, $n_+(T) = n_+(T - \varepsilon I)$, whose pivots are $\Delta_k(\varepsilon)/\Delta_{k-1}(\varepsilon)$ with leading minors Δ_k . The leading blocks are Jacobi matrices, with simple and strictly interlacing eigenvalues; if $\Delta_k(0) = 0$, counting their negative eigenvalues shows that $\Delta_k(\varepsilon)$ and $\Delta_{k-1}(0)$ have opposite signs. The recurrence $\Delta_{j+1} = t_{j+1}\Delta_j - a_j^2\Delta_{j-1}$ ($a_j \neq 0$ the off-diagonal entries) gives $\Delta_{k+1}(0) = -a_k^2\Delta_{k-1}(0)$ and $\Delta_{k+2}(0) = t_{k+2}\Delta_{k+1}(0)$. So $d_k(\varepsilon) \rightarrow 0^-$, $d_{k+1}(\varepsilon) \rightarrow +\infty$, $d_{k+2}(\varepsilon) \rightarrow t_{k+2}$. For n_- use $T + \varepsilon I$. $\square$

Lemma 5.3. *If $v_0, \dots, v_{d-1}$ are assigned, contributing Θ_d to $\Theta(G)$ and W to $n-b$, and $r = b-d$, then every completion with $n \leq N$ has $\Theta_d + 2r \leq \Theta(G) \leq \Theta_d + g(N - b - W) + 2(r-1)$, where $g(w) = (2+w)(1+w)$.*

Proof. A vertex of weight $w = p+2q$ contributes $g(w-q) + 2q \leq g(w)$, as $g(t) - g(t-1) = 2t+2$, and at least $g(0) = 2$. As g is convex, $g(a) + g(c) \leq g(a+c) + g(0)$, and g is increasing. $\square$

A node is skipped if no admissible value of $\Theta(G)$ from (1) lies in this window. The window is not monotone in p_k , so it never ends a loop.

5.3 Results, controls and versions

Table 1 lists the searches; runs split into parts are summed. Besides these, A (version 4, 15,678 nodes), C (no pruning, 1,601,329 sequences) and D (no pruning, all 3,031,469 sequences) found no integral generalized sun graph with cycle length 3 and at most 49 vertices.

N	b	A	B, B2	C; D
41	3	v4: 5,033 / 1,357	v1: 10,269 / 1,754	521,056 (np)
	4	v4: 398,136 / 85,505	v1: 410,544 / 52,641	5,683,267 (np); D: 410,544 (all)
	5, 7, 9	v4: no spectrum	v1: no spectrum	20,292,905; 82,161,850; 4,716,966
	8	v4: 70,518,560 / 576,976		57,561,471; D: 2,156,331
	10	v4: 50,716,095 / 52,652	B2 v1: 52,652 / 14,662	3,814,116; D: 2,124,182
	11, 13, 15		v1: no spectrum	$b = 11$: 1,239,693; $b = 13$: 0
	12	v4: 5,860,901 / 902		677,237; D: 618,755
42	3	v4: 5,334 / 1,381	v1: 11,895 / 2,030; v3: 3,991 / 2,030	606,367 (np); D: 1,246,783 (np)
	6	v4: 35,443,626 / 1,128,560; v6: 11,623,493 / 572,032	v1: 14,215,675 / 1,187,991; v3: 1,046,629 / 520,217; B2 v2: 1,128,560 / 449,500	201,673,369; D: 153,225,129 (all)
	10	v5: 465,028,393 / 461,049; v6: 19,679,948 / 71,680	v3: 401,175 / 200,579	C2: 4,601,920; D: 119,245,995 (all)
	14	v5: 904,115 / 850	v3: 0	1,951,786; D: 1,128,742
	18	v5: 18 / 0	v2: 2,048,976 / 56,952; v3 and B2 v2: 0	D: 67,817
	5–15 odd	v4 ($b \leq 9$): no spectrum	v1, v3: no spectrum	
	22, 26	v6: no spectrum		D: 0

Table 1: The searches with $n \leq N$. For A: search-tree nodes / exact leaf tests; for B and B2: sequences passing the moment tests / canonical sequences tested exactly; for C and D: sequences tested exactly, (np) meaning without pruning and (all) without symmetry reduction. The runs of D marked (np) have no symmetry reduction either, and its other runs test canonical sequences only. “No spectrum”: no admissible spectrum exists; for D, “0”: no sequence passes its pruning. Every run found nothing, except at $b = 4$ (A, B, C and D: the three graphs of Theorem 2) and at $b = 6$, $N = 42$ (every run: $C_{6,1}(0, 6, 6, 12, 6, 6)$ only). Version 6 of A used Lemma 4.2.

Controls and cross-checks. A and B also agree at $b = 4$, $N = 36$. Version 2 of B2 ran to completion at $b = 6$, $N = 42$ and found only $C_{6,1}(0, 6, 6, 12, 6, 6)$; the 1,128,560 sequences passing its moment tests are as many as the exact leaf tests of A (version 4). A note appended to its log says that this run was stopped, but the log also has the final line, which B2 prints only on completion, and a rerun gave the same final line. A run of B2 at $b = 10$, $N = 42$ was stopped unfinished and is not used. Version 6 of A at $b = 4$, $N = 49$ gives the same list with and without Lemma 4.2, including $C_{4,2}(10, 10, 0, 0)$ with 44 vertices and $\text{mult}(\pm 1) = 19$, and so does D. The first referee reran version 4 of A and version 1 of B2 at $b = 10$, $N = 41$: the sequences passing their moment tests form the same set of 52,652, none of them integral under the exact test of C. Without symmetry reduction, D found at $b = 6$, $N = 42$ exactly the six sequences of the dihedral orbit of $C_{6,1}(0, 6, 6, 12, 6, 6)$, and nothing at $b = 10$, $N = 42$. The numbers of sequences passing its pruning and of canonical sequences equal those of B (version 1) at $b = 4$, $N = 41$ (410,544 and 52,641) and at $b = 6$, $N = 42$ (14,215,675 and 1,187,991). Against exact characteristic polynomials (SymPy), Lemma 2.2, Lemma 2.3(a) and the bound of Lemma 4.4(a) hold on 2,968 random graphs ($b \leq 14$, $n \leq 60$; 1,551 with $\delta_0 > 0$, 953 with $\delta_1 + \delta_{-1} > 0$), and the polynomials of C agree modulo $2^{61} - 1$ on 400 graphs; the multiplicities of A agree with floating-point eigenvalues on 3,877 graphs, and the bounds of D equal the exact multiplicities

(nullities over $\mathbb{Q}$) on 3,250 random graphs ($b \leq 12$, $n \leq 60$). The second referee also rebuilt versions 5 and 6 of A and version 3 of B from their hashed sources and reproduced the logged counters of five of their runs. Two errors in earlier versions of A were found by these controls and by self-review, and corrected before the runs of Table 1: the first version used the path on all b cycle vertices, not a principal submatrix, and missed $C_{6,1}(0, 6, 6, 12, 6, 6)$; version 3 ended loops when the window of Lemma 5.3 failed. The superseded logs are kept.

Versions. The programs, logs, test scripts and both referee reports are in version 1.6.0 of [5], directory `certificates/sun-graphs/`, with A and B in `code/`, C in `review/code/` and D in `review/paper7/code/`; Table 2 lists the versions. The first referee agent audited versions 4 of A and 1 of B and B2, which produced all their runs with $N \leq 41$ and the runs of A and B with $N = 42$ for $b = 3, 5, 6, 7, 9$ (A) and $b \leq 15$ (B). The later versions, Lemma 4.2 and Corollary 4.3 were written after its report. The second referee agent then checked the two proofs line by line, read the sources of versions 5 and 6 of A, versions 2 and 3 of B and version 2 of B2, reproduced counters of the later versions as described above, and recomputed the cases of Table 3 marked D with implementation D.

file	SHA-256, first 16 digits	runs
<code>sunsearch_v4_f5a38c28.c</code>	<code>f5a38c28cf395886</code>	A v4: $N \leq 41$; $N = 42$, $b = 3, 5, 6, 7, 9$; $N = 49$, $b = 3$
<code>sunsearch_v5_bfcca593.c</code>	<code>bfcca593dc3ea0b2</code>	A v5: $N = 42$, $b = 10, 14, 18$
<code>sunsearch_v6_39a87195.c</code>	<code>39a871959d2b3d92</code>	A v6: $N = 42$, $b = 6, 10, 22, 26$; $N = 49$, $b = 4$
<code>sunB_v1_4d92eb48.c</code>	<code>4d92eb484cefea14</code>	B v1: $N \leq 42$, $b \leq 15$
<code>sunB_v2_8311d32e.c</code>	<code>8311d32e74f0cfbd</code>	B v2: $N = 42$, $b = 18$
<code>sunB.c</code> (version 3)	<code>f0eb3d9ca506a3f7</code>	B v3: $N = 42$
<code>sunB2_v1_54531ad0.c</code>	<code>54531ad075d1015f</code>	B2 v1: $N = 41$, $b = 10$
<code>sunB2.c</code> (version 2)	<code>c80538f9d4359954</code>	B2 v2: $N = 42$, $b = 6, 18$
<code>sunC.c</code>	<code>9401061cc17eff93</code>	C
<code>sunC2.c</code>	<code>b3e35c7bc97529ac</code>	C2: $N = 42$, $b = 10$
<code>sunD_v1.c</code>	<code>167d5a43514c4ef9</code>	D v1: $N = 41$, $b = 4$; $N = 42$, $b = 6$
<code>sunD2.c</code>	<code>5d913b626f9dc685</code>	D v2: $N = 41$, $b = 4, 8$; $N = 42$, $b = 6, 10$; $N = 49$, $b = 4$
<code>sunD3.c</code>	<code>e1a42832a46bd85d</code>	D v3: $N = 41$, $b = 10, 12$; $N = 42$, $b = 14, 18, 22, 26, 30$
<code>sunD3_maxb16.c</code>	<code>18a172d9cb9e42da</code>	D v3, smaller arrays: $N = 42, 49$, $b = 3$

Table 2: Program versions. The full SHA-256 values are recorded with the logs.

6 Proofs of Theorems 1 and 2

Proof of Theorem 1. Let G be a counterexample with $n \leq 42$; by [2, Theorem 2.1] it carries only copies of P_1 and P_2 . Odd $b \geq 5$ is excluded by Corollary 3.1. If $b \equiv 2 \pmod{4}$, then $b \in \{6, 10, 14, 18\}$ by Lemma 4.1, and $b \notin \{14, 18\}$ by Corollary 4.3. So $b \in \{3, 6, 10\}$, and the searches with $N = 42$ give $G \cong C_{6,1}(0, 6, 6, 12, 6, 6)$, a counterexample by [2]. $\square$

Proof of Theorem 2. Let G be integral, not a cycle, with $n \leq 41$; by [2, Theorem 2.1] it carries only copies of P_1 and P_2 . By Corollary 3.1 and Lemma 4.1, $b \in \{3, 4, 6, 8, 10, 12\}$. The searches find nothing for $b \in \{3, 8, 10, 12\}$, only the graph on 42 vertices for $b = 6$, and exactly the three graphs for $b = 4$. $\square$

case	by hand	by computation
<i>Theorem 1: $n \leq 42, b \not\equiv 0 \pmod{4}$</i>		
$b = 3$		A v4; B v1, v3; C and D (no pruning)
$b = 6$		A v4, v6; B v1, v3; B2 v2; C; D
$b = 10$		A v5 (without Lemma 4.2), v6; B v3; C2; D
$b = 14$	Corollary 4.3; Lemma 4.4(c)	A v5; C; D
$b = 18$	Corollary 4.3; Lemma 4.4(a),(b)	A v5; B v2; D
odd $b \geq 5$	Corollary 3.1; for $b \geq 15$ also Lemma 4.4(a),(b)	A v4 ($b \leq 9$), B ($b \leq 15$): no spectrum
even $b \geq 22$	Lemma 4.1; Lemma 4.4(a),(b)	A v6 ($b = 22, 26$): no spectrum; D ($b = 22, 26, 30$): none
<i>Theorem 2: $n \leq 41$</i>		
$b = 4$		A v4; B v1; C (no pruning); D
$b = 8, 12$		A v4; C; D
$b = 10$		also A v4; B2 v1; C and D at $N = 41$
even $b \geq 14$	Lemma 4.1; Lemma 4.4(a),(b)	
odd $b \geq 5$	Corollary 3.1; for $b \geq 15$ also Lemma 4.4(a),(b)	C for $b \leq 13$, without Corollary 3.1

Table 3: Confirmations of each case. A and B, with B2, were written separately by the agent that found the results; C, with C2, and D were written from scratch by two referee agents. The cases $b = 3, 6, 10$ of Theorem 2 are covered by those of Theorem 1.

7 Open questions

The questions of [2, Section 5], whether the cycle length b of an integral generalized sun graph other than a cycle must be even, and whether 4 and 6 are the only possible values, remain open. By Corollary 3.1, an odd b is 3, or at least 15 with $n \geq 102$; and $b = 3$ needs $n \geq 50$ (by the searches of Section 5). Lemma 4.4 bounds b for either parity: if $r_{\max}(n)$ is the largest t with $\sigma(t) \leq 2n$, then $b \leq r_{\max}(n) + 4$; writing $r_{\max}(n) = 4k + j$ with $0 \leq j \leq 3$, we have $2n \geq \sigma(4k) \geq 4 \sum_{v=1}^k v^2 \geq \frac{4}{3}k^3$, so $b \leq (96n)^{1/3} + 7$.

Thanks are due to Rodrigo O. Braga, Jean Carlo Moraes and Matheus C. Santos for the question answered here.

Tool and computational resource disclosure

This work was carried out on 26 September 2026 (UTC) in a workflow directed by the author, using AI coding agents: Anthropic Claude Code, with the model Claude Opus 5.5. Claude Code agents selected the problem, found the results and their proofs, wrote the programs, carried out the literature searches described in the introduction, and drafted this text. An independent Claude Code agent, which had not produced the results, refereed the proofs line by line, checked the lemmas of Section 2 against exact computations, compared the leaf sets of two programs, and reran the searches with its own program (implementation C), written from scratch; its corrections have been applied. A second referee agent, which had produced neither the results nor implementation C, later checked Lemma 4.2, Corollary 4.3 and the later program versions, and recomputed the cases $b = 3, 4, 6, 8, 10, 12, 14, 18$ with a program of its own (implementation D), written from scratch (Section 5); its corrections have been applied as well. Throughout, “we” refers to this workflow; the reading and searching described in the paper were done by the agents. The searches ran on one computer (macOS, arm64), one core per run; no completed run took more than an hour. The author is not a professional mathematician. Nothing in this paper is formally verified: the results rest on the written proofs and on the computations of Section 5. AI systems are not authors of this paper; the author takes full responsibility for its content.

References

- [1] R. O. Braga, R. R. Del-Vecchio and V. M. Rodrigues, Integral unicyclic graphs, *Linear Algebra Appl.* **614** (2021), 281–300, <https://doi.org/10.1016/j.laa.2020.05.002>.
- [2] R. O. Braga, J. C. Moraes and M. C. Santos, Pendant paths and integral generalized sun graphs, arXiv:2609.28754v1, 23 September 2026.
- [3] R. O. Braga, V. M. Rodrigues and V. Trevisan, Locating eigenvalues of unicyclic graphs, *Appl. Anal. Discrete Math.* **11** (2017), 273–298, <https://doi.org/10.2298/AADM1702273B>.
- [4] A. E. Brouwer and W. H. Haemers, *Spectra of Graphs*, Universitext, Springer, New York, 2012, <https://doi.org/10.1007/978-1-4614-1939-6>.
- [5] H. Dong, *ai4math-results*, repository, <https://github.com/Horace-Maxwell/ai4math-results>, version 1.6.0 (2026); archived at Zenodo under the concept DOI <https://doi.org/10.5281/zenodo.22970254>.
- [6] F. Harary and A. J. Schwenk, Which graphs have integral spectra?, in *Graphs and Combinatorics*, Lecture Notes in Math. 406, Springer, Berlin, 1974, 45–51, <https://doi.org/10.1007/BFb0066434>.
- [7] E. V. Haynsworth, Determination of the inertia of a partitioned Hermitian matrix, *Linear Algebra Appl.* **1** (1968), 73–81, [https://doi.org/10.1016/0024-3795\(68\)90050-5](https://doi.org/10.1016/0024-3795(68)90050-5).
- [8] A. Meurer et al., SymPy: symbolic computing in Python, *PeerJ Comput. Sci.* **3** (2017), e103, <https://doi.org/10.7717/peerj-cs.103>.
- [9] G. R. Omid, On integral graphs with few cycles, *Graphs Combin.* **25** (2009), 841–849, <https://doi.org/10.1007/s00373-010-0913-1>.
- [10] A. J. Schwenk, Computing the characteristic polynomial of a graph, in *Graphs and Combinatorics*, Lecture Notes in Math. 406, Springer, Berlin, 1974, 153–172, <https://doi.org/10.1007/BFb0066438>.
- [11] The Omega Institute, *trureturing* repository, <https://github.com/the-omega-institute/trureturing>, accessed 26 September 2026.
- [12] E. R. van Dam and W. H. Haemers, An odd characterization of the generalized odd graphs, *J. Combin. Theory Ser. B* **101** (2011), 486–489, <https://doi.org/10.1016/j.jctb.2011.03.001>; also arXiv:1202.2300.
- [13] M. S. T. Veras, Buscando grafos unicíclicos integrais [Searching for integral unicyclic graphs], in *XXXIII Salão de Iniciação Científica*, Porto Alegre, Brazil, 2021 (in Portuguese), https://lume.ufrgs.br/bitstream/handle/10183/243963/Resumo_72921.pdf, accessed 26 September 2026.